

Title

Flood Risk Assessment and Drainage Improvement Plan for Manila City

Author

Department of Engineering and Public Works

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Abstract

This study evaluates the increasing frequency of urban flooding in Manila City during heavy rainfall. Field inspections, rainfall records, drainage system assessments, and community surveys were conducted to identify the primary causes of flooding.

The assessment found that clogged drainage systems, inadequate pumping stations, improper waste disposal, and aging drainage infrastructure significantly contribute to flooding in several districts.

The study recommends regular drainage maintenance, expansion of pumping stations, installation of smart flood monitoring sensors, stricter enforcement of waste management policies, and rehabilitation of outdated drainage systems.

Implementing these recommendations is expected to reduce flood incidents, improve public safety, protect infrastructure, and minimize economic losses during the rainy season.